

POWERFACTORY

PowerFactory 2024

Technical Reference

CT Adapter

RelCtadapt, TypCtadapt

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document describes different configurations and the corresponding dynamic and steady-state behaviour of the *CT Adapter* unit for protection devices in *PowerFactory*.

CT adapters are used to compensate different voltage levels, CT ratios or CT connections for differential protection. The corresponding real world equivalents are auxiliary current transformers used for electromechanical or static protection devices or the numerical compensation logic in digital devices.

1.1 CT Adapter Type (TypCtadapt)

The *CT Adapter Type* defines the general characteristic and available ranges. The *Type* selection specifies the number of input and output signals available. The *Reference Adapter* selection is used to define the reference system for the differential protection and to form groups of *CT Adapters*. Please refer to Section 1.3 for additional information. For *3-phase* adapters a configuration for zero-sequence current elimination is also available.

The nominal voltage and current transformer ratio ranges are optional, but this should be consistent within a group, meaning that either all the *CT Adapter Types* should have ranges entered, or none.

1.2 CT Adapter (RelCtadapt)

The *CT Adapter* is used to specify the actual settings of an adapter. The input fields correspond to the ranges available in the type. In addition a *current transformer connection* and the corresponding *phase rotation* can be entered for *3-phase* adapters. Please refer to Section 2.3 for additional information.

1.3 Connection Schemes

This section shows some typical connection schemes for differential protection applications and how to form the adapter groups. A group of CT adapters is formed by assigning the *CT Adapter Type* of the intended reference to the “Reference Adapter” field of all other *CT Adapter Types* in the group. The “Reference Adapter” field of the reference itself remains empty. A group of CT adapters can only have one reference to ensure that all output signals have the same frame of reference. It is recommended to connect the reference adapter to the differential protection as the first set of inputs, so that the differential protection uses the reference to form its internal reference system.

1.3.1 2-Winding Transformer and Lines

The following connection is the most common one and can be used for differential protection of 2-winding transformers. Although less common, it can also be used for line differential protection to compensate different CT ratios on both line ends. As shown in Figure1.1 one adapter per side is required. In most cases the HV side CT will be used as reference adapter, so the phase shift of the transformer is entered on the LV side adapter.

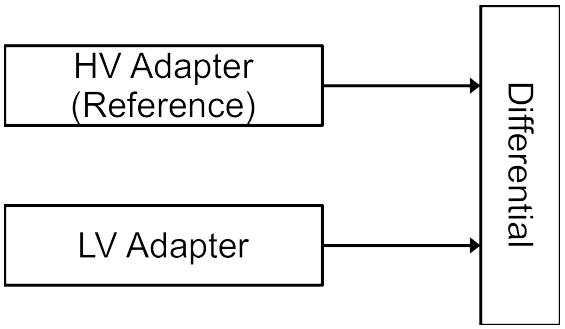


Figure 1.1: Example connection for a 2-winding transformer or line protection

1.3.2 3-Winding Transformer

The connection for 3-winding transformer, shown in Figure 1.2 is very similar to the 2-winding transformer connection, but with one additional adapter. The HV side is commonly used as a reference and both MV and LV adapters refer to the HV adapter. The phase shift between the HV and MV windings is entered in the MV adapter and the phase shift between the HV and LV winding entered in the LV adapter.

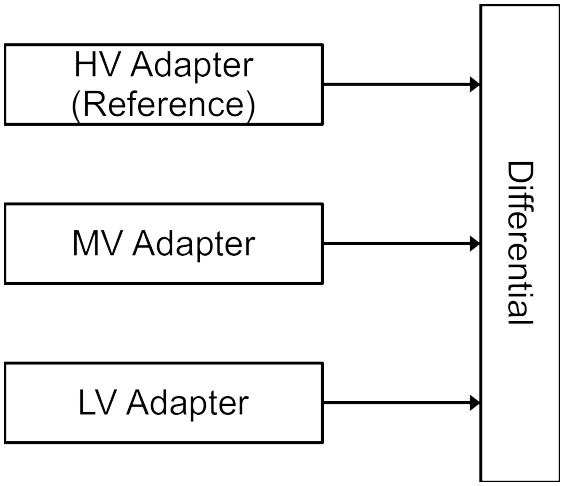


Figure 1.2: Example connection for a 3-winding transformer protection

1.3.3 Restricted Earth Fault

Another common differential protection is the restricted earth fault which combines a 3-phase adapter for the phase currents with a single phase adapter for the star point current. This adapter configuration is mostly used to compensate for different CT ratios, as both CTs are located on the same side of the transformer.

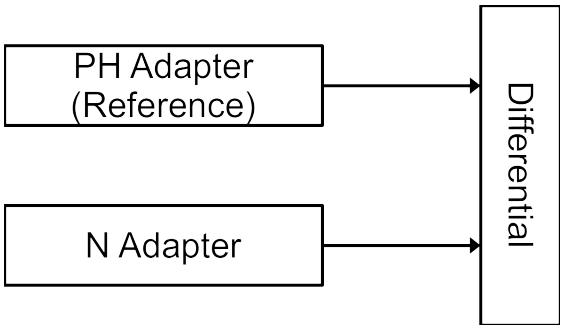


Figure 1.3: Example connection for a restricted earth fault protection

2 Calculation Models

This section describes the equations used for 3- and 1-phase CT adapters. For readability the equations are shown with complex phasors, as these are used in the majority of calculations within *PowerFactory*, but the equations are also valid when used with instantaneous values for input and output.

2.1 3-phase

$$\mathbf{I}_{out} = \frac{U}{U_{ref}} \cdot \frac{t}{t_{ref}} \cdot \mathbf{C} \cdot \mathbf{I}_{in} \tag{1}$$

Where:

$\mathbf{I}_{out}$ is the vector of the output currents in sec.A (*s:Ir_X* $+ j \cdot$ *s:Ii_X*)

$$\mathbf{I}_{out} = \begin{pmatrix} \underline{I}_{out,A} \\ \underline{I}_{out,B} \\ \underline{I}_{out,C} \end{pmatrix}$$

$\mathbf{I}_{in}$ is the vector of the input currents in sec.A (*s:wIr_X* $+ j \cdot$ *s:wIi_X*)

$$\mathbf{I}_{in} = \begin{pmatrix} \underline{I}_{in,A} \\ \underline{I}_{in,B} \\ \underline{I}_{in,C} \end{pmatrix}$$

$\mathbf{C}$ is the compensation matrix (see Section 2.3)

U**,** U_{ref} are the nominal voltages of the adapter and reference adapter respectively (*e:LLVolt*)

t**,** t_{ref} are the CT ratios of the adapter and reference adapter respectively (*e:CTratio*)

If the CTs have different vector groups, the resulting amplitude error is compensated by an additional factor of $\frac{1}{\sqrt{3}}$, if the reference adapter is *delta*, or $\sqrt{3}$ if the adapter is *delta*.

Note: For the reference adapter itself the following applies: $U = U_{ref}$, $t = t_{ref}$ and $\mathbf{C} = \mathbf{C}_0$

If *Remove earth current* is enabled the input currents $\mathbf{I}_{in}$ are modified as follows:

$$\underline{I}_0 = (\underline{I}_{in,A} + \underline{I}_{in,B} + \underline{I}_{in,C})/3 \tag{2}$$

$$\mathbf{I}_{in} = \begin{pmatrix} \underline{I}_{in,A} - \underline{I}_0 \\ \underline{I}_{in,B} - \underline{I}_0 \\ \underline{I}_{in,C} - \underline{I}_0 \end{pmatrix} \tag{3}$$

Where:

$\underline{I}_{in,X}$ is the input current for phase X in sec.A (*s:wIr_X* $+ j \cdot$ *s:wIi_X*)

2.2 1-phase

$$\underline{I}_{out} = \frac{U}{U_{ref}} \cdot \frac{t}{t_{ref}} \cdot \underline{I}_{in} \tag{4}$$

Where:

$\underline{I}_{out}$ is the vector of the output currents in sec.A (*s:Ir_X* $+ j \cdot$ *s:Ii_X*)

$\underline{I}_{in}$ is the vector of the input currents in sec.A (*s:wIr_X* $+ j \cdot$ *s:wIi_X*)

U, U_{ref} are the nominal voltages of the adapter and reference adapter respectively (*e:LLVolt*);

t, t_{ref} are the CT ratios of the adapter and reference adapter respectively (*e:CTratio*)

Note: For the reference adapter itself the following applies: $U = U_{ref}$, $t = t_{ref}$

2.3 Compensation Matrices

The compensations matrices are used to compensate the phase shift caused by transformer vector groups or different CT connections. The phase shift is always relative to the *reference* adapter and includes the transformer phase shift and a potential phase shift from CT connections. The corresponding correction matrix for a (total) phase shift of $n * 30°$ is shown as $\mathbf{C}_n$ below.

$$\mathbf{C}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{C}_1 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \quad \mathbf{C}_2 = \frac{1}{3} \cdot \begin{bmatrix} 1 & -2 & 1 \\ 1 & 1 & -2 \\ -2 & 1 & 1 \end{bmatrix}$$

$$\mathbf{C}_3 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \quad \mathbf{C}_4 = \frac{1}{3} \cdot \begin{bmatrix} -1 & -1 & 2 \\ 2 & -1 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

$$\mathbf{C}_5 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \quad \mathbf{C}_6 = \frac{1}{3} \cdot \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

$$\mathbf{C}_7 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \mathbf{C}_8 = \frac{1}{3} \cdot \begin{bmatrix} -1 & 2 & -1 \\ -1 & -1 & 2 \\ 2 & -1 & -1 \end{bmatrix}$$

$$\mathbf{C}_9 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \quad \mathbf{C}_{10} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & -2 \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}$$

$$\mathbf{C}_{11} = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad \mathbf{C}_{12} = \frac{1}{3} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

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